

1. Administrative & Study Identification

Full Study Title: A Study of Opevesostat (MK-5684) Versus Alternative Next-generation Hormonal Agent (NHA) in Metastatic Castration-resistant Prostate Cancer (mCRPC) Post One NHA (MK-5684-004) **Official Title:** MK-5684-004: A Phase 3, Randomized, Open-label Study of Opevesostat Versus Alternative Abiraterone Acetate or Enzalutamide in Participants With Metastatic Castration-resistant Prostate Cancer (mCRPC) That Progressed On or After Prior Treatment With One Next-generation Hormonal Agent (NHA) (OMAHA-004) **Short Title/Acronym:** OMAHA-004 / MK-5684-004 **Protocol Number:** MK-5684-004 **NCT Number:** NCT06136650 **Sponsor Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** Opevesostat (MK-5684) **Phase of Development:** Phase 3 **Trial Status:** Active, not recruiting **Medical Monitor:** MSD Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether opevesostat is superior to an alternative next-generation hormonal agent (abiraterone acetate or enzalutamide) with respect to radiographic progression-free survival (rPFS) in androgen receptor ligand-binding domain (AR LBD) mutation-positive and -negative participants. **Secondary Objectives:** To evaluate Overall Survival (OS), Time to Initiation of the First Subsequent Anticancer Therapy (TFST), Objective Response Rate (ORR), Duration of Response (DOR), time to pain progression, time to PSA progression, and time to first symptomatic skeletal-related event (SSRE).

2.2 Background & Rationale To address resistance to standard androgen receptor pathway inhibitors (ARPIs) in patients with metastatic castration-resistant prostate cancer (mCRPC). Resistance is often associated with AR-LBD mutations allowing promiscuous stimulation by other steroid hormones. Opevesostat is an oral nonsteroidal inhibitor of CYP11A1, targeting the first and rate-limiting step of steroid biosynthesis, effectively suppressing all steroid hormones that fuel prostate cancer progression.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Abiraterone acetate or Enzalutamide) **Blinding:** Open-label **Randomization:** 1:1, Stratified by presence versus

absence of AR-LBD mutations **Phase:** Phase 3 **Trial Status:** Active, not recruiting

3.2 Planned Enrollment & Duration Target **NS:** 1,314 adult male patients **Number of Sites:** Global multicenter trial **Estimated Study Start:** 18-Dec-2023 **Estimated Primary Completion:** 10-May-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years, male. 2. Histologically or cytologically confirmed adenocarcinoma of the prostate without small cell histology. 3. Prostate cancer progression while receiving androgen deprivation therapy (ADT) (or post-bilateral orchiectomy) within 6 months before screening, with ongoing ADT maintaining serum testosterone < 50 ng/dL. 4. Disease progression during or after treatment with exactly 1 next-generation hormonal agent (NHA/ARPI) for at least 8 weeks (at least 14 weeks if bone progression) for HSPC or CRPC. Note: Prior docetaxel for HSPC is permitted if ≤ 6 cycles with no radiographic disease progression during treatment. 5. Current evidence of distant metastatic disease (M1) documented by bone scan and/or CT/MRI. 6. Has had prior treatment with a poly (ADP-ribose) polymerase inhibitor (PARPi), or deemed ineligible to receive PARPi, or refused PARPi treatment. 7. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 assessed within 10 days before randomization. 8. Provided tumor tissue from a fresh core or excisional biopsy from soft tissue not previously irradiated (samples from tumors progressing at a prior radiation site are allowed). 9. Controlled viral infections permitted: Hepatitis B (HBsAg positive) allowed if on antiviral therapy ≥ 4 weeks with undetectable viral load; Hepatitis C allowed if viral load is undetectable; HIV allowed if well-controlled on antiretroviral therapy (ART). 10. Adequate organ function and recovery from prior therapy AEs to $\leq$ Grade 1 (or baseline). $\leq$ Grade 2 neuropathy/osteopenia allowed.

4.2 Exclusion Criteria 1. Prior taxane-based chemotherapy for mCRPC. 2. Received > 1 NHA or prior targeted small molecule therapy within 4 weeks of the first dose. 3. "Superscan" appearance on bone scan. 4. Active central nervous system (CNS) metastases or carcinomatous meningitis (radiologically and clinically stable treated brain metastases not requiring steroids for ≥ 14 days are allowed). 5. Presence of a gastrointestinal condition or inability to swallow capsules/tablets. 6. History of pituitary dysfunction or poorly controlled diabetes mellitus. 7. Clinically significant abnormal serum potassium or sodium levels. 8. Hypotension (systolic BP < 110 mmHg) or uncontrolled hypertension (systolic BP ≥ 160 mmHg or diastolic BP ≥ 90 mmHg). 9. History of active/unstable cardio/cerebrovascular disease, history or family history of long QTc syndrome, clinically significant ventricular arrhythmias, or heart blocks

without a pacemaker. 10. History of seizure(s) within 6 months before consent or conditions predisposing to seizures within 12 months. 11. Current treatment with CYP450-inducing antiepileptic drugs for seizures, or systemic use of strong CYP3A4 inducers and P-gp inhibitors within 2 weeks prior to first dose. 12. Prior radiotherapy within 2 weeks of first dose, or systemic anticancer therapy within 4 weeks prior to first dose. 13. Received a live or live-attenuated vaccine within 30 days before the first dose. 14. Concurrent active Hepatitis B virus and Hepatitis C virus infection, or active infection requiring systemic therapy. 15. Chronic systemic steroid therapy (> 10 mg daily of prednisone equivalent) or immunodeficiency.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Experimental Arm:** Opevesostat 5 mg twice daily (BID) + hormone replacement therapy (dexamethasone 1.5 mg once daily [QD] + fludrocortisone 0.1 mg QD). * **Active Comparator Arm:** Abiraterone acetate 1,000 mg QD + prednisone or prednisolone 5 mg BID (if prior enzalutamide) OR enzalutamide 160 mg QD (if prior abiraterone). **Route of Administration:** Oral **Duration of Treatment:** Until radiographic disease progression, unacceptable toxicity, investigator's decision, or consent withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Maintenance of ADT (LHRH agonist/antagonist) to keep testosterone at castrate levels. **Prohibited:** CYP450-inducing antiepileptic drugs, strong CYP3A4 inducers, P-gp inhibitors, live vaccines, and chronic systemic steroid therapy > 10 mg daily of prednisone equivalent.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Imaging, AR-LBD mutation status testing, fresh biopsy
Baseline	Day 0	Randomization, ECOG (within 10 days prior), First Dose
Interim	Every 4 Weeks	Safety Labs, Efficacy Assessment (PSA), HRT monitoring
Radiographic Imaging	Every 8-12 Weeks	CT/MRI and Bone Scans
End of Study	Variable	Final Vital Signs, Exit Interview, IP Accountability, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Radiographic Progression-Free Survival (rPFS). Time from randomization to the first documented disease progression or death due to any cause. **Measurement**

Method: Prostate Cancer Working Group 3 (PCWG3)-modified RECIST v1.1 assessed by Blinded Independent Central Review (BICR).

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) *(Note: Per Protocol Amendment 08, OS was moved to be a secondary outcome measure).* * Time to Initiation of the First Subsequent Anticancer Therapy (TFST) * Objective Response Rate (ORR) * Duration of Response (DOR) * Time to PSA progression and PSA50 response rate * Time to first symptomatic skeletal-related event (SSRE)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via regular site visits, monitoring for glucocorticoid/mineralocorticoid imbalances due to HRT. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (iDMC) to assess ongoing safety and efficacy.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcomes; Safety Set for AE/SAE evaluations. Endpoints analyzed separately for AR-LBD mutation-positive and -negative cohorts. **Sample Size Justification:** $N = 1,314$ participants adequately powered to detect superiority in rPFS and OS. **Handling of Missing Data:** Standard Kaplan-Meier censoring methods for time-to-event endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to ensure protocol adherence and HRT management. **Audit Trail:** eCRF locking and independent BICR evaluation of imaging parameters.

11. Study Identifiers & Governance

Posting ID: 478256242563842654 **Primary ID:** MK-5684-004 **Registry Number:** NCT06136650 **Sponsor/Lead Organization:** Merck **Study Title:** A Study of Opevesostat (MK-5684) Versus Alternative Next-generation Hormonal Agent (NHA) in Metastatic Castration-resistant Prostate Cancer (mCRPC) Post One NHA (MK-5684-004) **Official Regulatory Title:** MK-5684-004: A Phase 3, Randomized, Open-label Study of Opevesostat Versus Alternative Abiraterone Acetate or Enzalutamide in Participants With Metastatic

Castration-resistant Prostate Cancer (mCRPC) That Progressed On or After Prior Treatment With One Next-generation Hormonal Agent (NHA) (OMAHA-004)

12. Clinical Framework (mCRPC Specific)

Primary Condition: Metastatic Castration-Resistant Prostate Cancer (mCRPC) **Disease Areas / Neoplasms:** Metastatic Castration-resistant Prostate Cancer (mCRPC), Prostatic Neoplasms, Neoplasms **Diagnosis Threshold:** Prostate cancer progression on ADT with testosterone ≤ 50 ng/dL **Medical Methodology:** CYP11A1 Inhibition vs Androgen Receptor Pathway Inhibitor (ARPI) Switch **Optimization Target:** Complete suppression of steroidogenesis to prevent AR signaling activation

13. Study Procedures & Methodology

Management Pathway: Hormonal therapy switch in mCRPC post one NHA **Education Protocol (HCP):** Protocol training and BICR imaging guidelines **Education Protocol (Patient):** HRT compliance, dosing guidelines, and self-learning materials **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for rPFS and adverse event tracking

14. Observation & Follow-Up Schedule

Total Duration: Until radiographic disease progression or unacceptable toxicity **Onsite Visit Cadence:** Every 4 weeks **Remote Monitoring Frequency:** Every 4 weeks (Q4W) **Checklist Review Interval:** Imaging scans every 8 to 12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				